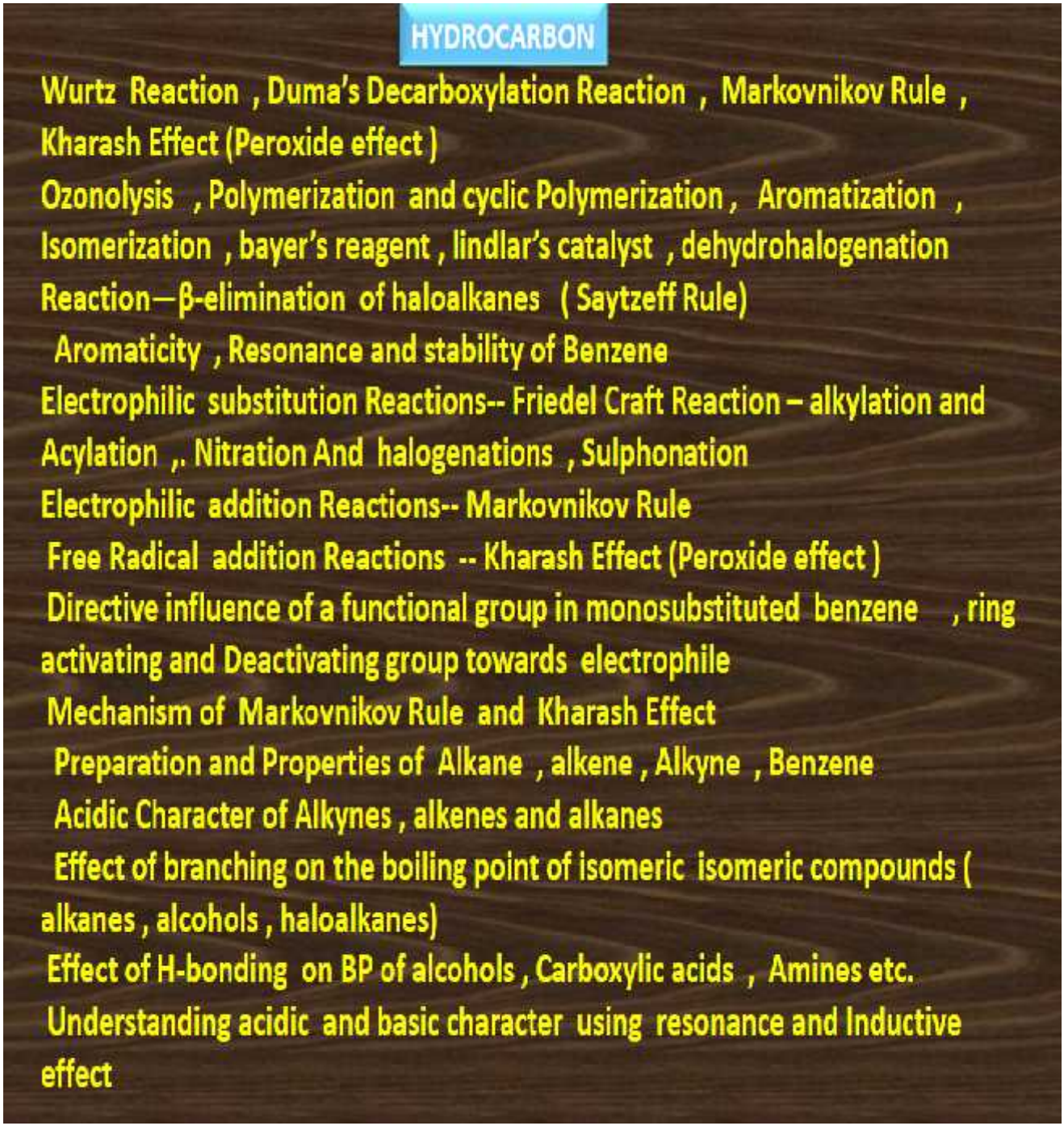
CHAPTER 13

HYDROCARBON

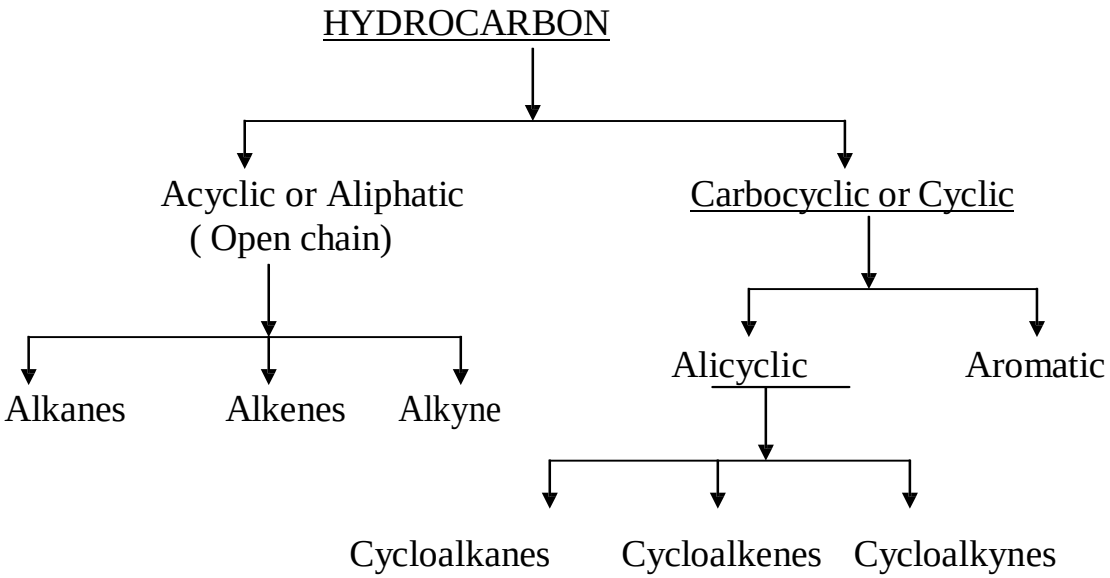


- Hydrocarbons are composed of Carbon and hydrogen.
- The important fuels like Petrol, kerosene, coal gas, CNG, LPG etc. are all hydrocarbons or their mixture.

Sources:

Petroleum and natural gas are the major sources of aliphatic hydrocarbon while coal is an important source of aromatic hydrocarbons. The oil trapped inside the rocks is known as petroleum. PETRA – ROCK, OLEUM – OIL. The oil in the petroleum field is covered with a gaseous mixture known as natural gas. The main constituents of the natural gas are methane, ethane, propane and butane.

CLASSIFICATION OF HYDROCARBONS:



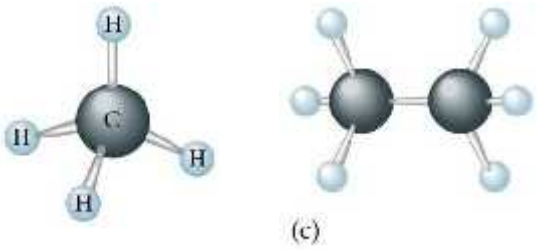
Depending upon the carbon-carbon bond present they are classified into

i) Saturated Hydrocarbons : Bonds present between carbon atoms and hydrogen atoms is single bond.
If carbon atoms form a closed chain or ring they are termed as cycloalkanes.

ii) Unsaturated Hydrocarbons: Contain carbon-carbon multiple bonds- double bonds, triple bonds or both.

iii) Aromatic Hydrocarbons

Alkanes:-



- Paraffins
- General formula C_nH_{2n+2}
- sp^3 hybridisation
- C–C bond length 1.15 4 A^0
- Chemically unreactive
- Show chain, position and optical isomerism.
- Heptane has 9 isomer, Octane 18 and Decane 75.

Nomenclature:

Preparation:-

❶ Wurtz reaction:-

$$2CH_3CH_2Br + 2Na \xrightarrow[\text{ether}]{\text{Dry}} CH_3CH_2CH_2CH_3 + 2NaBr$$

- Follow mainly free radical mechanism
- Useful in preparing an alkane containing even number of carbon atoms
- Stepping up reaction

❷From Grignard reagent (RMgX)

$$RMgX + HOH \rightarrow RH + Mg(OH)X$$
$$RMgX + R'OH \rightarrow RH + Mg(OR')X$$
$$RMgX + R'NH_2 \rightarrow RH + Mg(NHR')X$$

❸From unsaturated hydrocarbons:- Sabatier-Senderens reduction

$$R - CH = CH_2 + H_2 \xrightarrow{Ni/\Delta} R - CH_2 - CH_3$$

$$R - C \equiv CH + H_2 \xrightarrow{Ni/\Delta} R - CH_2 - CH_3$$

4. From carboxylic acids-

Decarboxylation.-

$$CH_3COO^-Na^+ + NaOH \xrightarrow{CaO/\Delta} CH_4 + Na_2CO_3$$

Kolbe's electrolytic method-

$$2CH_3COO^-Na^+ + 2H_2O \xrightarrow{Electrolysis} CH_3\text{-}CH_3 + 2CO_2 + H_2 + 2NaOH$$

❖ **Physical Properties:-**

(1) Nature:- Non-Polar due to covalent nature of C—C bond and C—H bond. C—C bond enrgy = 83 kj/mole and C—H bond energy = 99 kj/mole.
C_1—C_4 = gases, C_5—C_{17} = colourless odourless liquid and > C_{17} = Solid.

(2) Solubility:- Like dissolve like

Viz, Polar compounds dissolve in polar solvent and Non-Polar compound dissolve in non polar solvent.

(3) Boiling point:- Low boiling point due to non polar in nature.

The molecules are held together only by weak Van der Waalls' forces.

Since we known that the magnitude of Van der Waalls' forces is directly proportional to the molecular size. Therefore, the boiling point increases with increase the molecular size i.e. with increase in number of carbon atoms.

Noted:- the boiling points of the branched chain Alkanes are less than the straight chain isomers.

This is due to the fact that branching of the chain makes the molecule more compact and thereby decreases the surface area and consequently, the magnitudes of Van der Waalls' forces also decrease.

Example: Boiling point of pentane is 309.1 K and that of 2,2-dimethylpropane is 282.5 K.

❖ **Chemical properties**

1. **Combustion:**

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ $\quad \Delta_c H^\theta$ = -890kJ/mol

General Combustion Equation:

$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \rightarrow nCO_2 + (n+1)\,H2O$

During incomplete combustion carbon black is formed which is used in the manufacture of ink, printer ink, black pigments and as filters.

$$CH_4(g) + O_2(g) \xrightarrow{Incomplete\ combustion} C(s) + 2H_2O(l)$$

2. Oxidation:

i) $2CH_4 + O_2 \xrightarrow{Cu/523K/100atm} 2CH_3OH$

ii) $CH_4 + O_2 \xrightarrow{Mo_2O_3} HCHO + H_2O$

iii) $2CH_3\text{-}CH_3 + 3O_2 \xrightarrow{(CH_3COO)_2Mn} 2CH_3COOH + 2H_2O$

iv) $(CH_3)_3CH \xrightarrow{KMnO_4\ Oxidation} (CH_3)_3COH$

3. **Substitution Reaction:**

Halogenation:

$CH_4 + Cl_2 \xrightarrow{uv} CH_3Cl + HCl$

$CH_3Cl + Cl_2 \xrightarrow{uv} CH_2Cl_2 + HCl$

$CH_2Cl_2 + Cl_2 \xrightarrow{uv} CHCl_3 + HCl$

$CHCl_3 + Cl_2 \xrightarrow{uv} CCl_4 + HCl$

- Iodination is a reversible reaction. So it is carried out by heating alkane in the presence of some oxidizing agent like iodic acid (HIO3) or nitric acid (HNO3) or mercuric oxide (HgO) which oxidizes HI formed during the reaction.

  $CH_4 + I_2 \rightarrow CH_3I + HI$

  $HIO_3 + 5HI \rightarrow 3I_2 + 3H_2O$
- Fluorination of alkane takes place explosively resulting even in the rupture of C—C bond in higher alkanes.
- Features of Halogenations:-

(i) The reactivity of Halogens: $F_2 > Cl_2 > Br_2 > I_2$.

(ii) The rate of replacement of hydrogen of alkanes is: $3° > 2° > 1°$

Mechanism: Halogenation reactions take place by free radical mechanism. The reaction proceeds in the following steps:

(i) Chain Initiation:

$Cl\text{-}Cl \xrightarrow{hv} 2\dot{Cl}$

(ii) Chain Propagation:

$CH_4 + \dot{Cl} \xrightarrow{hv} C\dot{H}_3 + H\text{-}Cl$

$C\dot{H}_3 + Cl\text{-}Cl \xrightarrow{hv} CH_3\text{-}Cl + \dot{Cl}$

(iii) Chain Termination:

$\dot{Cl} + \dot{Cl} \rightarrow Cl\text{-}Cl$

$H_3\dot{C} + \dot{C}H_3 \rightarrow H_3C\text{-}CH_3$

$H_3\dot{C} + \dot{Cl} \rightarrow H_3C\text{-}Cl$

4. Aromatization:

$H_3C(CH_2)_4CH_3 \xrightarrow{Cr_2O_3, 773K/10-20\ atm} C_6H_6$

This method is also called dehydrogenation or hydroforming. Similarly, heptane gives toluene, n-Octane give o-xylene and 2, methyl heptane give m-xylene.

5. Pyrolysis: When higher alkanes are heated at high temp (about 700-800k) in the presence of alumina or silica catalysts, the alkanes break down to lower alkanes and alkenes.

$C_6H_{14} \xrightarrow{773K} C_3H_6 + C_2H_4 + CH_4$

6. Action of steam: - catalyst- nickel, alumina Al_2O_3

$CH_4 + H_2O\ (steam) \xrightarrow{Ni, 1000℃} CO + 3H_2$

7 Isomerisation:-

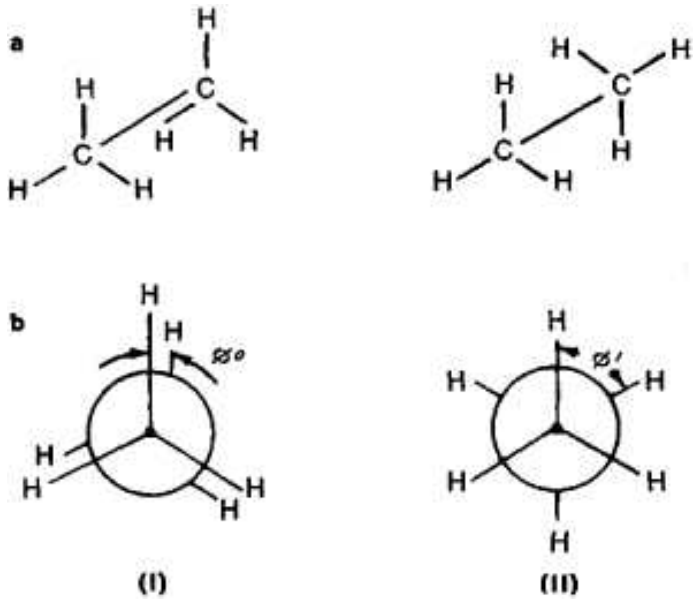
$CH_3(CH_2)_4CH_3 \xrightarrow{Anhyd.AlCl_3/HCl} CH_3\text{-}CH(CH_3)\text{-}(CH_2)_2\text{-}CH_3 + CH_3\text{-}CH_2\text{-}CH(CH_3)\text{-}CH_2\text{-}CH_3$

- **CONFORMATIONAL ISOMERISM:**

The different molecular arrangements arising as a result of rotation around carbon carbon single bonds are called conformational isomers or rotational isomers and the phenomenon is called conformational isomerism.

Numerous possible arrangements of ethane are possible. Two extreme conformations are known. These are eclipsed conformation and staggered conformation.

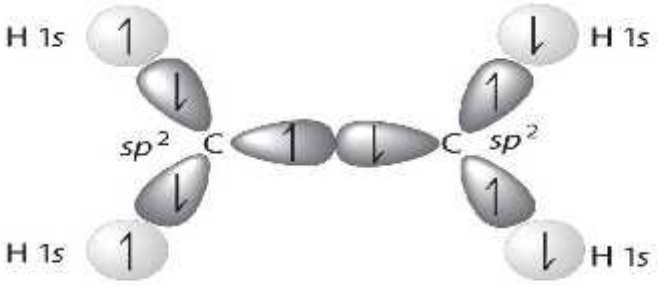
SAWHORSE AND NEWMAN REPRESENTATION



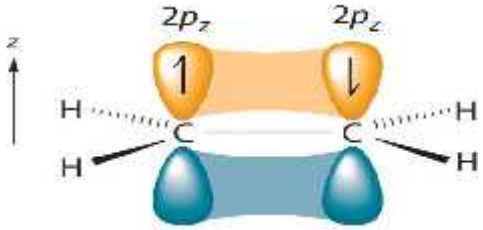
ALKENES

- Unsaturated hydrocarbon which have double bond.
- General molecular formula C_nH_{2n}
- C–C bond hybridization 1.34 A^0
- sp^2 hybridization
- When we treated Alkene with chlorine, oily products are obtained. So Alkenes are also known as Olefins. (Greek olefiant meaning oil forming).
- Show chain, positional and geometrical isomerism

Structure of double bond:-



(a) C_2H_4 σ-bonded framework



(b) C_2H_4 π bonding

❖ **Preparation:-**

1. From Alkynes:- Alkynes on partial reduction with Partially deactivated palladised charcoal known as *Lindlar's catalyst* give alkenes.

$$\underset{\text{Ethyne}}{CH\equiv CH} + H_2 \xrightarrow{Pd/C} \underset{\text{Ethene}}{CH_2=CH_2}$$

2. From Haloalkanes: - Dehydrohalogenation ($\beta-$ elimination)

$$CH_3\text{-}CH_2Cl \xrightarrow{\text{alco.KOH}/\Delta} CH_2=CH_2 + HCl$$

3. From Dihaloalkanes(Vicinal Dihalides):- Dehalogenation

$$CH_2Br\text{-}CH_2Br + Zn \rightarrow CH_2=CH_2 + ZnBr_2$$

4. From alcohols by acidic dehydration:

$$CH_3\text{-}CH_2\text{-}OH \xrightarrow{\text{conc.}H_2SO_4,\Delta} CH_2=CH_2 + H_2O$$

Chemical Properties:-

Addition Reaction:- Alkene show electrophilic addition reaction.

1. Addition of Hydrogen:-

$$R\text{-}CH=CH_2 + H_2 \xrightarrow{Ni} R\text{-}CH_2\text{-}CH_3$$

2. Addition of Halogens:-

$CH_2=CH_2 + Br_2 \xrightarrow{CCl_4} CH_2Br\text{-}CH_2Br$

The reddish orange colour of bromine solution in carbon tetrachloride is discharged when bromine adds up to an unsaturation site. This reaction is used as a test for unsaturation.

3. Addition of hydrogen halides:-

Addition reaction of HBr to symmetrical alkenes

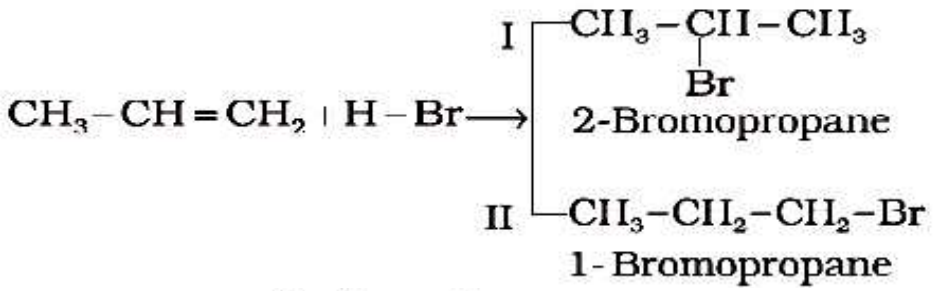
$CH_2=CH_2 + HBr \rightarrow CH_3CH_2Br$

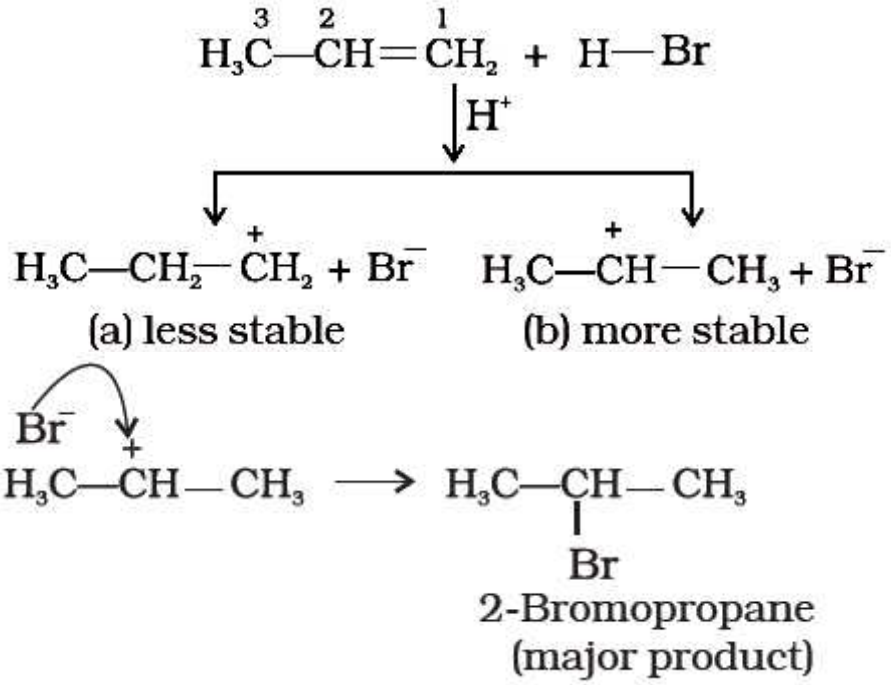
Addition reaction of HBr to unsymmetrical alkenes takes place according to Markovnikov Rule.

Markownikov Rule:- Negative part of the addendum (adding molecule) gets attached to that carbon atom which possesses lesser number of hydrogen atoms.

$CH_3CH=CH_2 + HBr \rightarrow CH_3CHBr\text{-}CH_3$

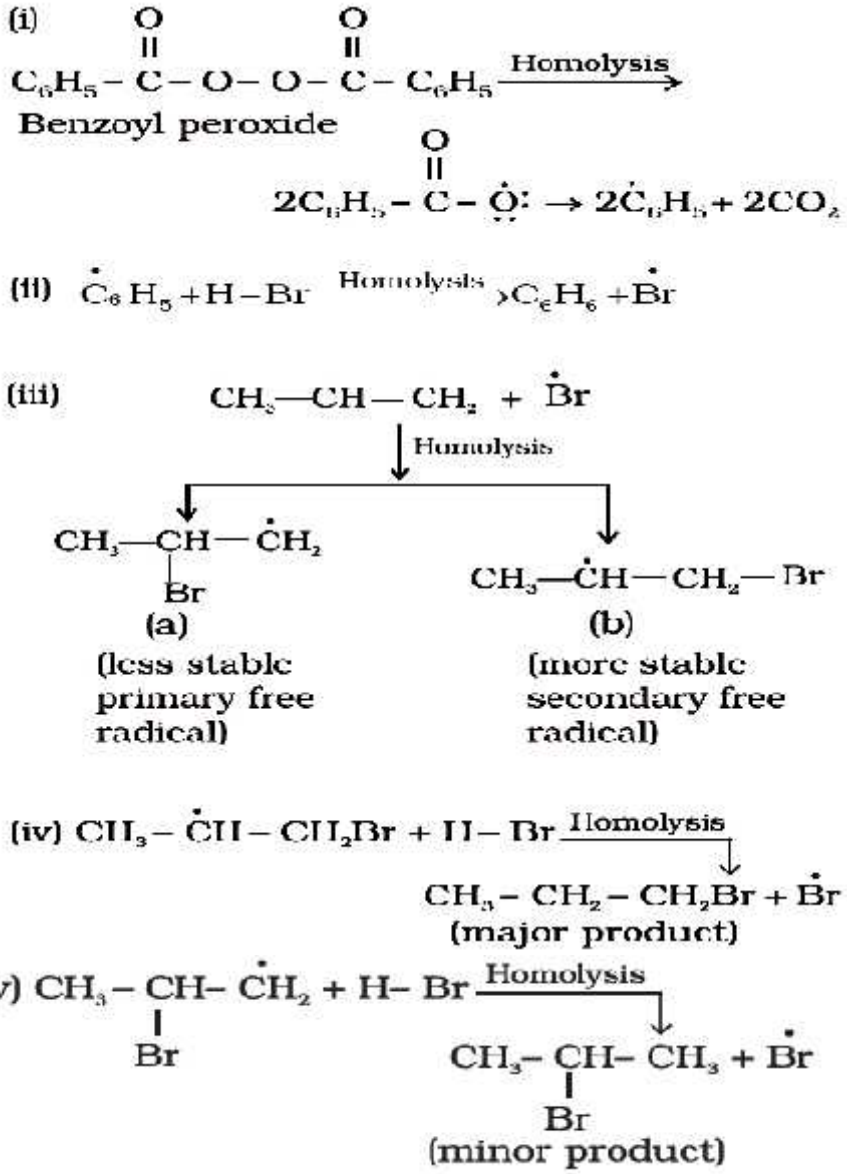
MECHANISM





Peroxide effect or Kharash (Anti Markownikov's addition):- In 1933 Kharash and Mayo observed that when HBr is added to an unsymmetrical double bond in the presence of organic peroxide, the reaction take places opposite to the Markovnikov rule.

$$CH_3\text{-}CH{=}CH_2 + HBr \xrightarrow{(C_6H_5CO)_2O_2} CH_3\text{-}CH_2\text{-}CH_2\text{-}Br$$



Noted:- peroxide effect is applicable only to HBr and not to HF, HCl and HI. Addition of HF, HCl and HI takes place according to Markovnikov's rule even in the presence of peroxide.

4. Addition of Sulphuric Acid:

$$CH_2{=}CH_2 + H_2SO_4 \rightarrow C_2H_5HSO_4$$

5. Addition of water:

$$CH_2{=}CH_2 + H_2O \xrightarrow{H^+} CH_3\text{-}CH_2\text{-}OH$$

6. Oxidation:

$CH_2=CH_2 + H_2O + O \xrightarrow{dil.KMnO_4,\ 273K} CH_2OH\text{-}CH_2OH$

Noted:- The alkaline potassium permanganate solution is known as Baeyer's reagent. It has bright pink colour. It oxidizes alkenes to glycols which is colourless. This reaction is used as a test for the presence of double bond in a molecule. This is also known as Baeyer test.

$(CH_3)_2\text{-}C=CH_2 \xrightarrow{KMnO_4/H^+} (CH_3)_2C=O + CO_2 + H_2O$

$CH_3\text{-}CH= CH\text{-}CH_3 \xrightarrow{KMnO_4/H^+} 2CH_3COOH$

7. Ozonolysis:

$CH_3CH=CH_2 + O_3 \xrightarrow{Zn+H_2O} CH_3CHO + HCHO$

Noted:- Bromine water test and Baeyer's test are used to detect the presence of double bond while ozonolysis is used to detect the position of double bond.

8. Polymerisation:

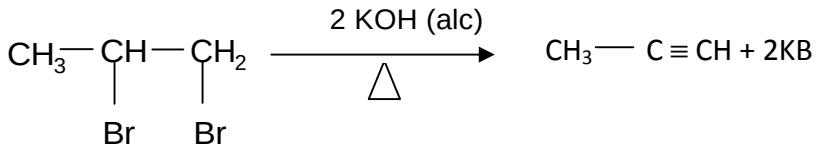
$n(CH_2=CH_2) \xrightarrow{\text{High temp./pressure,catalyst}} -(CH_2\text{-}CH_2)_n\text{-}$

Alkynes

- ☞ Unsaturated hydrocarbon which have triple bond.
- ☞ General molecular formula C_nH_{2n-2}
- ☞ sp hybridization
- ☞ Shows chain, positional and functional isomerism

❖ Preparation:-

From vicinal dihalides: - dehalogenation



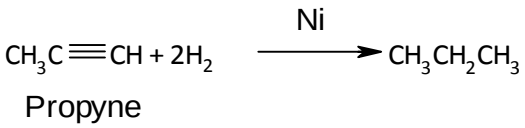
By the action of water on calcium carbide:-

$CaC_2+H_2O \rightarrow HC \equiv CH+Ca(OH)_2$

❖ Chemical Properties:-

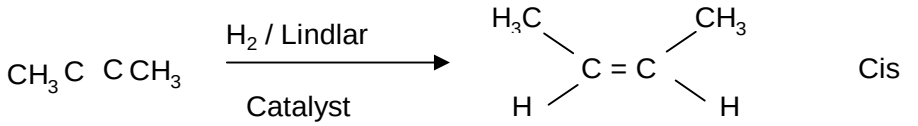
❶ Addition Reaction:- Alkyne show electrophilic addition reaction.

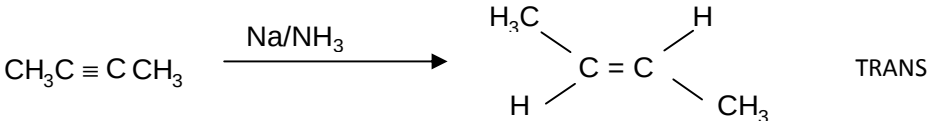
① Addition of Hydrogen:- Hydrogenation.



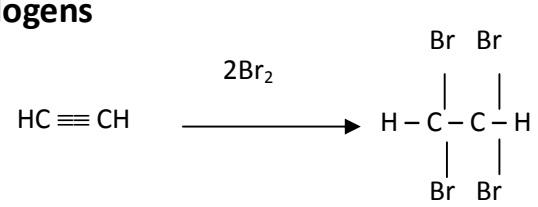
Noted:- It may be noted that the hydrogenation can be controlled at the alkene stage only. This is possible by using a *Lindlar's catalysts* or *sodium in liquid NH_3* at 200k temp..

Noted:- It may be again noted that the catalytic reduction of alkynes in the presence of Lindlar's catalyst gives cis-alkenes while in the presence of sodium in liquid NH_3 (Birch reduction) gives trans-alkenes.





Addition of Halogens



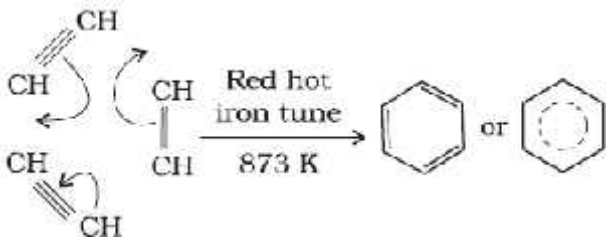
Addition of hydrogen halides

$$HC \equiv CH + 2HBr \rightarrow CH_3CHBr_2$$

$$CH_3C{\equiv}CCH_2 \xrightarrow[HgSO_4]{H_2O/H^+} CH_3CH_2\overset{\overset{O}{\|}}{C}CH_2CH_3\overset{\overset{O}{\|}}{C}CH_2CH_2CH_3$$

Polymerisation-

a. ***Linear polymerisation*:** of ethyne gives polyacetylene or polyethyne which is a high molecular weight polyene containing repeating units of (CH = CH – CH = CH) and can be represented as —(CH = CH – CH = CH)n —
b. Cyclic polymerization- results in the formation of aromatic compound.



Acidity of Alkynes- Terminal alkynes are acidic in nature.

$$HC \equiv CH + Na \rightarrow HC \equiv C^{-}Na^{+} + \tfrac{1}{2}H_2$$

Monosodium ethynide

$$CH_3 - C \equiv C - H \quad + \quad Na^{+}NH_2^{-}$$

$$\downarrow$$

$$CH_3 - C \equiv C^{-}Na^{+} \quad + \quad NH_3$$

Sodium propynide

Alkanes, alkenes and alkynes follow the following trend in their acidic behaviour :

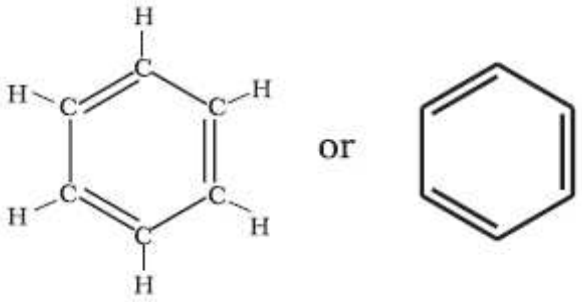
i) $HC \equiv CH > H_2C = CH_2 > CH_3 - CH_3$

ii) $HC \equiv CH > CH_3 - C \equiv CH >> CH_3 - C \equiv C - CH_3$

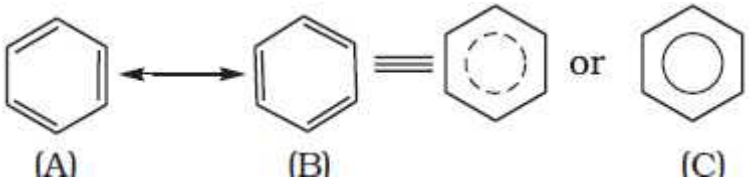
AROMATIC HYDROCARBON

Aromatic compounds containing benzene ring are known as **benzenoids** and those not containing a benzene ring are known as **non-benzenoids**.

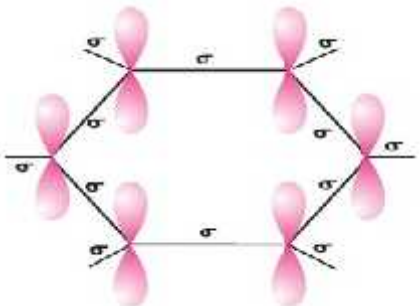
Structure of Benzene- Kekul*é* structure



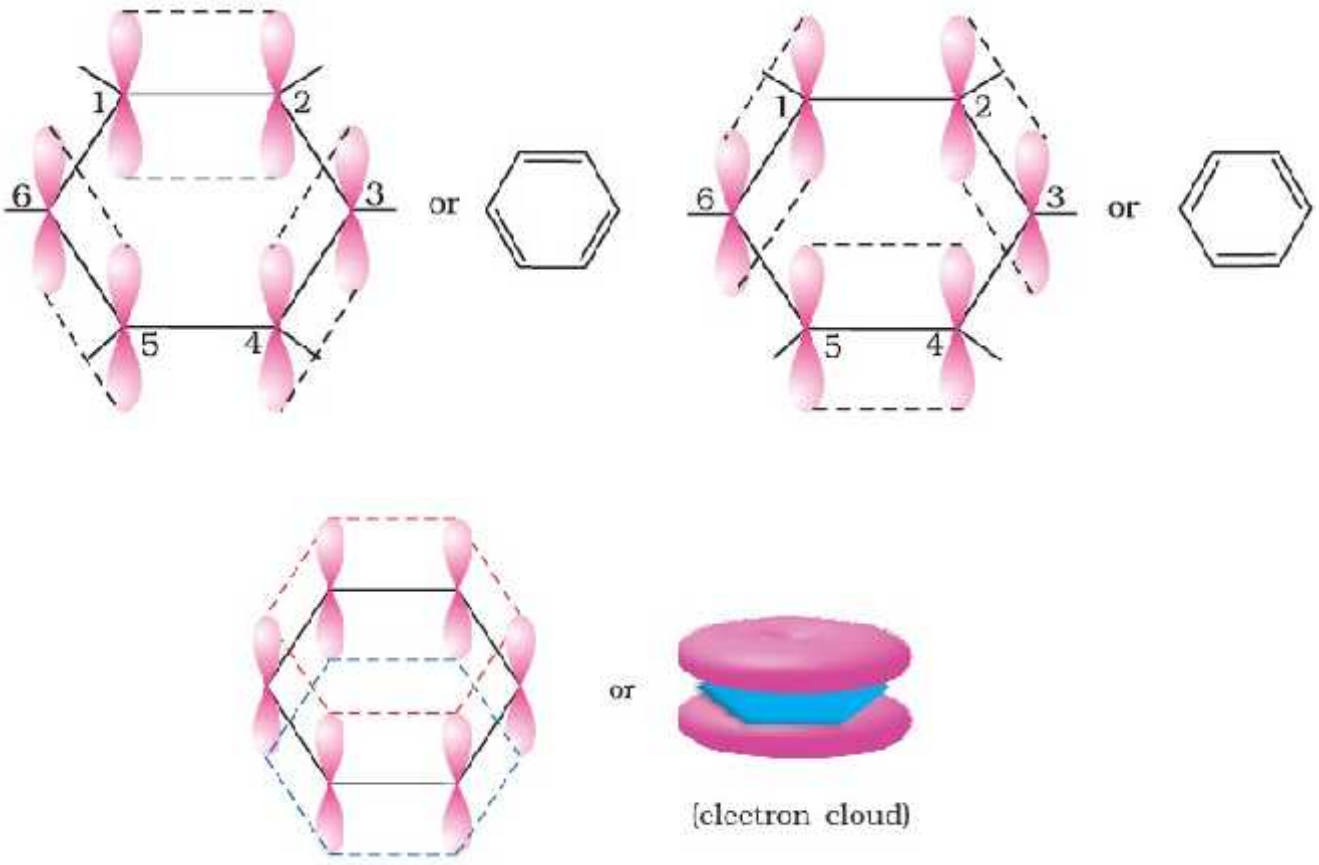
Resonance and stability of benzene-Benzene is a hybrid of various resonating structures.



The orbital overlapping picture benzene- All the six carbon atoms in benzene are sp^2 hybridized and these hybrid orbitals form sigma bonds.



The unhybridised *p* orbital of carbon atoms are close enough to form a π bond by lateral overlap.



The six π electrons are thus delocalised and can move freely about the six carbon nuclei. The delocalised π electron cloud is attracted more strongly by the nuclei of the carbon atoms than the electron cloud localized between two carbon atoms. Therefore, presence of delocalised π electrons in benzene makes it more stable .

Aromaticity:- The compounds that follow the following features are to be considered aromatic.

(i) Planarity

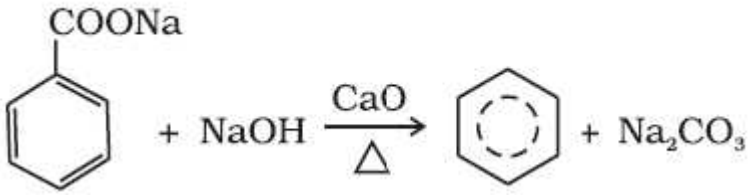
(ii) Complete delocalisation of the π electrons in the ring

(iii) Presence of $(4n + 2)$ π electrons in the ring where n is an integer (n = 0, 1, 2) This is often referred to as **Hückel Rule**.

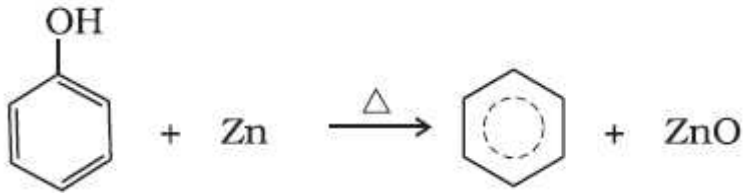
Preparation of Benzene:

(i) Cyclic polymerisation of ethyne:

(ii) Decarboxylation of aromatic acids:



(iii) ***Reduction of phenol:*** Phenol is reduced to benzene by passing its vapours over heated zinc dust



Physical properties:

1. Aromatic hydrocarbons are non- polar molecules and are usually colourless liquids or solids with a characteristic aroma.
2. Aromatic hydrocarbons are immiscible with water but are readily miscible with organic solvents.
3. They burn with sooty flame.

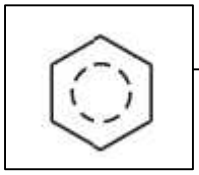
Chemical properties

Arenes are characterised by electrophilic substitution reactions proceed via the following three steps:

(a) Generation of the eletrophile

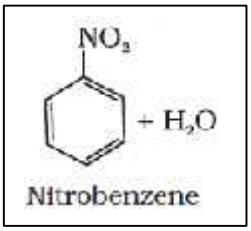
(b) Formation of carbocation intermediate

(c) Removal of proton from the carbocation intermediate



Nitration

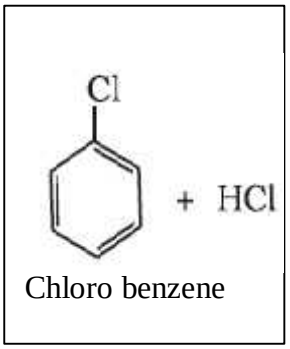
Conc.HNO_3+Conc.H_2SO_4



Halogenation

+Cl_2

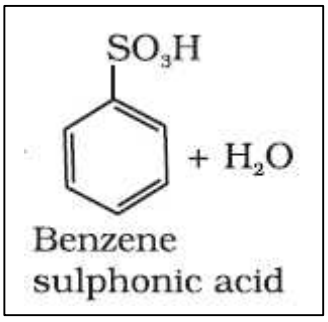
Anhyd. $AlCl_3$



Sulphonation

Fuming sulphuric acid

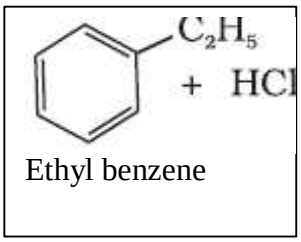
$H_2SO_4(SO_3)$



Friedel-Crafts alkylation

+C_2H_5Cl

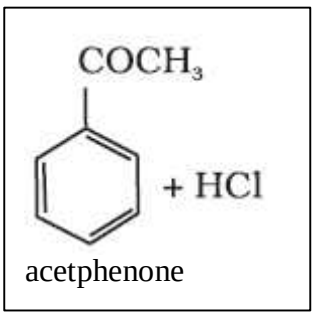
Anhyd. $AlCl_3$



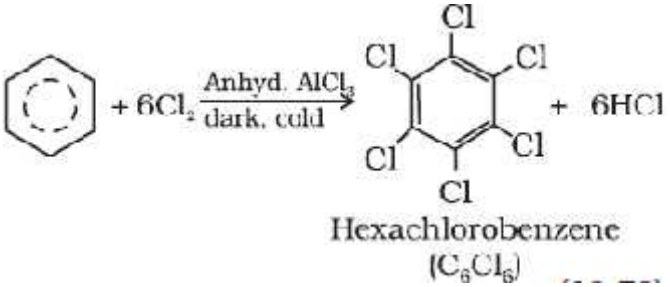
Friedel-Crafts acylation

+CH_3COCl

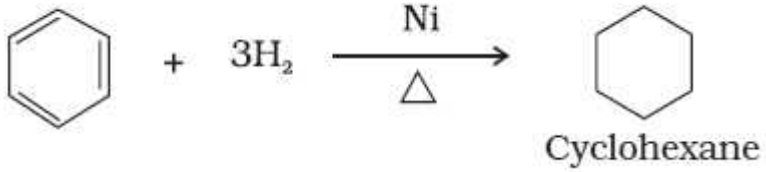
Anhyd. $AlCl_3$

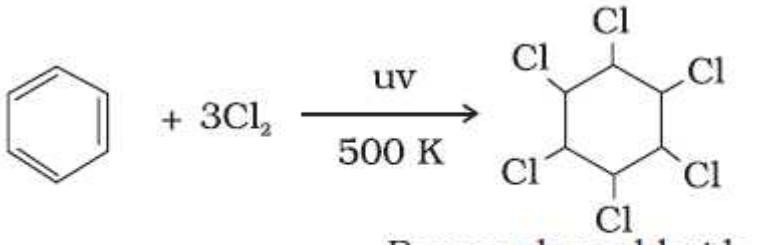


Benzene on treatment with excess of chlorine in the presence of anhydrous $AlCl_3$ in dark yields hexachlorobenzene (C_6Cl_6) (Substitution reaction)



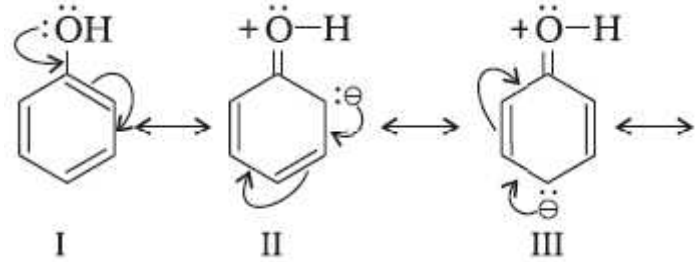
Addition reactions of benzene-



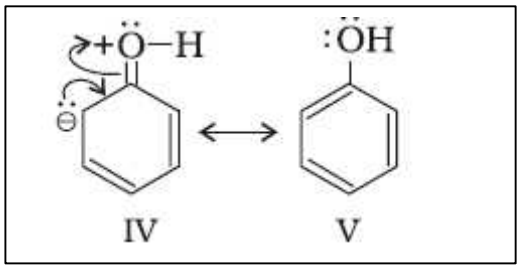


Directive influence of a functional group in monosubstituted benzene:-

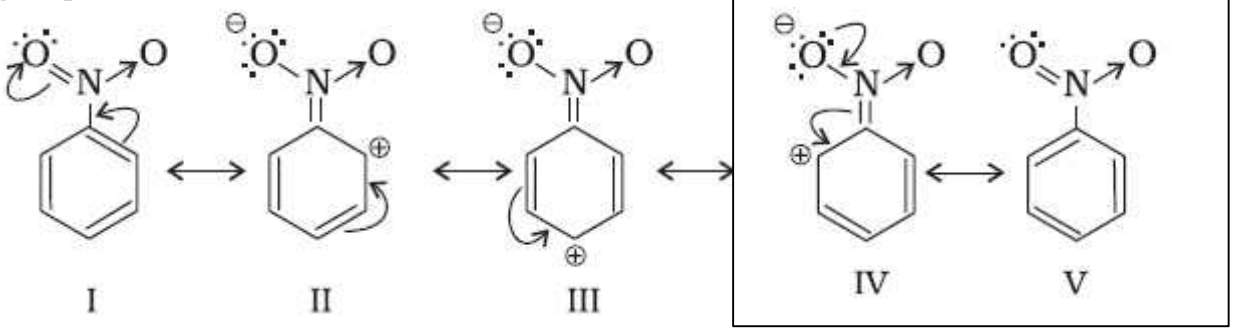
1. Ortho and para directing groups and activating- –OH, $–NH_2$, –NHR, – $NHCOCH_3$, $–OCH_3$, $–CH_3$, $–C_2H_5$, etc.



2. *Meta directing group and deactivating:*–NO_2, –CN, –CHO, –COR, –COOH, – $COOR$, $–SO_3H$, etc.

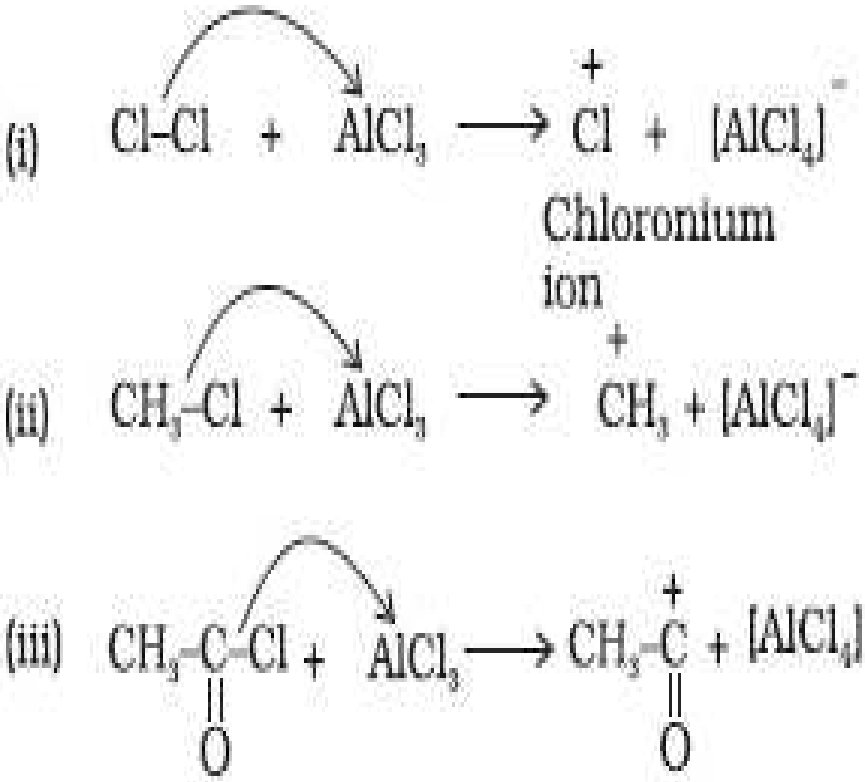


3. Ortho and para directing groups and deactivating- Halogens because of their strong – I effect, overall electron density on benzene ring decreases. However, due to resonance the electron density on *o*– and *p*– positions is greater than that at the *m*-position. Hence, they are also *o*– and *p*– directing groups.

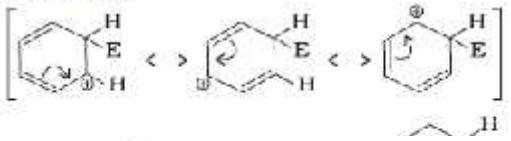


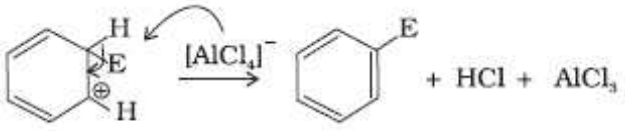
IONS.

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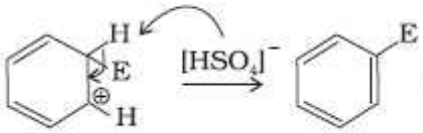


The arenium ion gets stabilised by resonance:

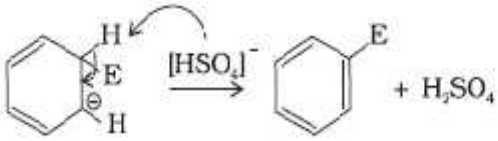




ermediate



Cl3



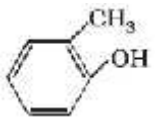
CARCINOGENICITY AND TOXICITY-Benzene and polynuclear hydrocarbons containing more than two benzene rings fused together are toxic and said to possess cancer producing (carcinogenic) property.

ONE MARK QUESTIONS

1. **What are hydrocarbons?**
   Ans. Compounds of hydrogen and carbon.
2. **What is the general formula of alkanes?**
   Ans. C_nH_{2n+2}
3. **Write the general formula of alkenes.**
   Ans. C_nH_{2n}
4. **What is the general formula of alkynes?**
   Ans. C_nH_{2n-2}
5. **Give the IUPAC name of lowest molecular weight alkane that contains a quaternary carbon.**
   Ans. 2,2dimethylpropane.
6. **Arrange the following in the increasing order of C-C bond length- C_2H_6 C_2H_4 C_2H_2**
   Ans. C_2H_2 < C_2H_4 < C_2H_6
7. **Out of ethylene and acetylene which is more acidic and why?**
   Ans. Acetylene, due to greater electonegativity of the sp hybridised carbon.
8. **Name two reagents which can be used to distinguish between ethene and** ethyne.
   Ans.Tollen's reagent and ammonical CuCl solution.
9. **Arrange the following in order of decreasing reactivity towards alkanes. HCl, HBr, HI, HF**
   Ans.HI> HBr> HCl >HF
10. **How will you detect the presence of unsaturation in an organic compound?**
    Ans. Generally Unsaturated organic compound decolourise Bayer's reagent and Bromine water.
11. **What is Grignard reagent?**
    Ans. Alkyl magnesium halides (RMgX)

TWO MARKS QUESTIONS

1. **Write the IUPAC names of the following-**

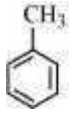
   a. $CH_2=CH-C\equiv C-CH_3$ b. 

   Ans. a .Pent-en-3-yne b. 2-methylphenol

2. **Write chemical equations for combustion reaction of (i) Butane (ii) Toluene** Ans.

(i) $2C_4H_{10(g)} + 13O_{2(g)} \longrightarrow 8CO_{2(g)} + 10H_2O_{(g)} + \text{Heat}$

Butane

(ii)  $+ 9O_{2(g)} \longrightarrow 7CO_{2(g)} + 4H_2O_{(g)} + \text{Heat}$

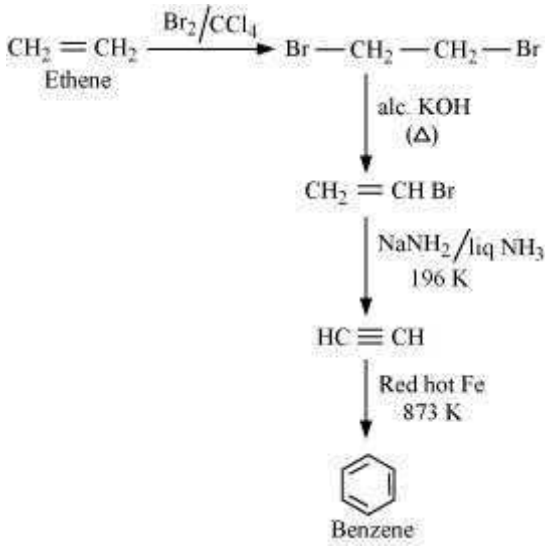
Toluene

3. **What are the necessary conditions for any system to be aromatic?**
Ans. A compound is said to be aromatic if it satisfies the following three conditions: **(i)** It should have a planar structure.
(ii) The π–electrons of the compound are completely delocalized in the ring.
(iii)The total number of π–electrons present in the ring should be equal to $(4n + 2)$, where $n = 0, 1, 2 \ldots$ etc. This is known as Huckel's rule.
4. **What effect does branching of an alkane chain has on its boiling point?**
Ans. As branching increases, the surface area of the molecule decreases which results in a small area of contact. As a result, the Van der Waals force also decreases which can be overcome at a relatively lower temperature. Hence, the boiling point of an alkane chain decreases with an increase in branching.
5. **How would you convert the following compounds into benzene?**
(i) Ethyne (ii) Ethene

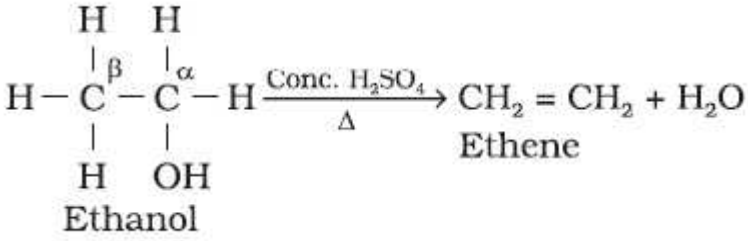
Ans. **(i)** Benzene from Ethyne:



(ii) Benzene from Ethene:



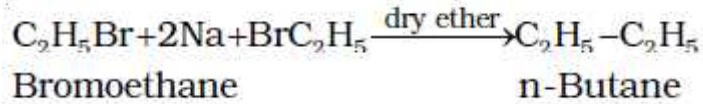
6. **Suggest the name of Lewis acids other than anhydrous aluminium chloride which can be used during ethylation of benzene.**
   Ans. anhydrous $FeCl_3$, $SnCl_4$, BF_3 etc.
7. **Write the name of all the possible isomers of $C_2H_2Cl_2$ and indicate which of them is non-polar.**
   Ans.(i) cis-1,2-dichloroethene (ii) trans-1,2-dichloroethene
   (iii) 1,1- dichloroethene. trans-1,2-dichloroethene is non-polar.
8. **Although benzene is highly unsaturated, it does not undergo addition reactions, why?**
   Ans. Because of extra stability due to delocalization of π-electrons.
9. **What are alkanes? Why are they called paraffins?**
   Ans. Those hydrocarbons which contain single bond between carbon- carbon are called alkanes. They are called paraffins because they are very less reactive (Latin- Parum= little, affins = affinity)
10. **How can ethene be prepared from (i) ethanol (ii) ethyl bromide?**
    Ans. (i) Ethene from ethanol- by acidic dehydration of alcohols



(ii) Ethene from ethyl bromide- by dehydrohalogenation of ethyl bromide
$CH_3CH_2Br + KOH\ (alc) \rightarrow H_2C = CH_2 + KBr + H_2O$

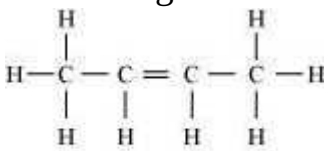
THREE MARKS QUESTIONS

1. **What is Wurtz reaction? How can it be used to prepare butane?**
   Ans- When alkyl halides is treated with metallic Na in presence of dry ether, alkanes are formed. This reaction is called Wurtz reaction.
   Butane is prepared by the reaction of bromoethane with metallic Na in presence of dry ether



2. **An alkene 'A' contains three C – C, eight C – H σ bonds and one C – C π bond. 'A' on ozonolysis gives two moles of an aldehyde of molar mass 44 u. Deduce IUPAC name of 'A'.**
   Ans.. The formation of two moles of an aldehyde indicates the presence of identical structural units on both sides of the double bond containing carbon atoms. Hence, the structure of 'A' can be represented as:
   XC = CX
   There are eight C–H σ bonds. Hence, there are 8 hydrogen atoms in 'A'. Also, there are three C–C bonds. Hence, there are four carbon atoms present in the structure of 'A'.

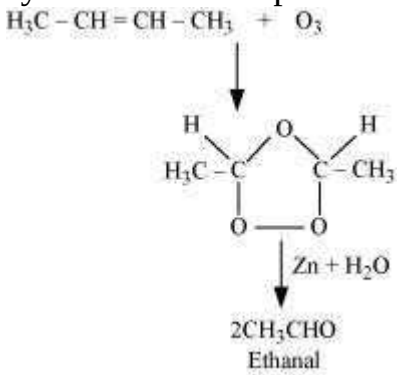
Combining the inferences, the structure of ‘A’ can be represented as:



(A)

the IUPAC name of ‘A’ is But-2-ene.

Ozonolysis of ‘A’ takes place as:



The final product is ethanal with molecular mass

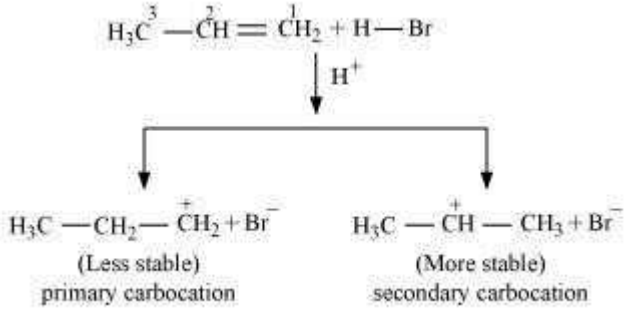
$$=\left[(2\times12)+(4\times1)+(1\times16)\right]$$
$$=44\text{ u}$$

FIVE MARKS QUESTIONS

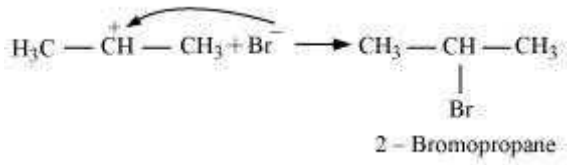
1. **Addition of HBr to propene yields 2-bromopropane, while in the presence of benzoyl peroxide, the same reaction yields 1-bromopropane. Explain and give mechanism**

Ans. Addition of HBr to propene is an example of an electrophilic substitution reaction.

Hydrogen bromide provides an electrophile, H^+. This electrophile attacks the double bond to form 1° and 2° carbocations as shown:

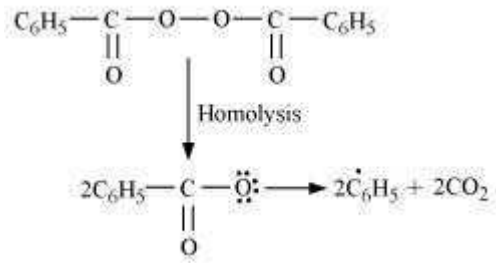


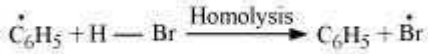
Secondary carbocations are more stable than primary carbocations. Hence, the former predominates since it will form at a faster rate. Thus, in the next step, Br^- attacks the carbocation to form 2 – bromopropane as the major product.

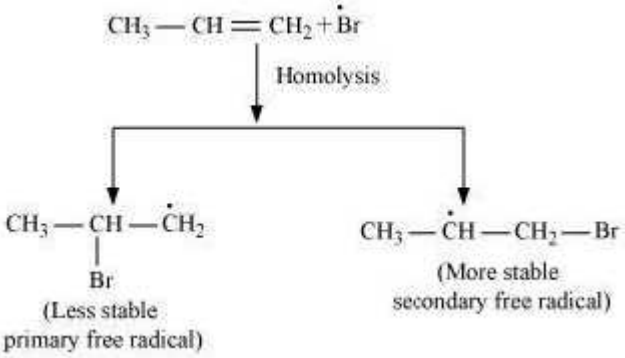


This reaction follows Markovnikov's rule

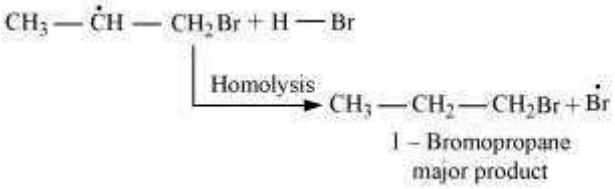
In the presence of benzoyl peroxide, an addition reaction takes place anti to Markovnikov's rule. The reaction follows a free radical chain mechanism as:







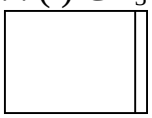
Secondary free radicals are more stable than primary radicals. Hence, the former predominates since it forms at a faster rate. Thus, 1 – bromopropane is obtained as the major product.



HOTS QUESTIONS

1. Give the expected products when the following compounds are subjected to oxidation by hot $KMnO_4$ solution.
(i) But-2-ene
(ii) Cyclobutene

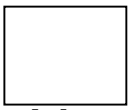
Ans: . (i) CH_3-CH=CH-CH_3 $\xrightarrow{KMnO_4}$ 2 CH_3COOH

(ii) 

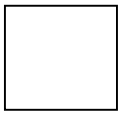
Cyclobutene $\xrightarrow{KMnO_4}$ HOOC-CH_2-CH_2-COOH

2. A hydrocarbon C_4H_8 neither decolourises bromine dissolved in CCl_4 nor reacts with HBr. When heated to 473K with hydrogen in the presence Ni catalyst, a new hydrocarbon C_4H_{10} is formed. What is the original hydrocarbon?

Ans: The molecular formula of the hydrocarbon represents the following four structural isomers.

$CH_3CH_2CH=CH_2$	$CH_3CH=CHCH_3$	$(CH_3)_2C=CH_2$	
But-1-ene	But-2-ene	2-Methylpropene	Cyclobutane

Since the hydrocarbon does not decolourise bromine dissolved in CCl_4 and also fails to react with HBr; this means that it is not unsaturated in nature. The possibility of being an alkene is ruled out. Therefore it must be cyclobutane which reacts with hydrogen to form butane.

 + H_2 $\xrightarrow{Ni\ ,473K}$ $CH_3CH_2CH_2CH_3$

3. A saturated hydrocarbon was found to have vapour density 36.It forms only single monochlorosubstituted product. Suggest its name.

Ans: Molecular mass of hydrocarbon=2xV.D=2x36=72.
Since hydrocarbon is saturated: C_nH_{2n+2}=72 or

$$12n + 2n + 2 = 72 \text{ or } n=5$$

Molecular formula of hydrocarbon = C_5H_{12}

Since it forms only one monochloro derivative it must be symmetrical.
The hydrocarbon is

$$(CH_3)_3CCH_3 + Cl_2 \rightarrow (CH_3)_3CCH_2Cl + HCl$$

4. Why do the C-C bonds break rather than the C-H bonds during pyrolysis of alkanes?
Ans: Bond energy of C-C bond (83.0 kcal/mole) is lower than bond energy of C-H bond(98.8 kcal/mole).Thus C-C bonds break more easily C-H bonds.

5. Methane does not react with chlorine in dark. Explain by giving a reason.
Ans: Chlorination of methane is a free radical substitution. To convert Cl_2 molecules into chlorine free radicals, energy is required which is not available in dark. Thus methane does not react with Cl_2 in dark due to absence of free radicals.

6. Complete the following equations:
(i) $CH_3CH_2C{\equiv}CH_3 \xrightarrow{\text{Cold dilute } KMnO_4}$ **?**
(ii) $CH_3CH_2C{\equiv}CCH_3 \xrightarrow{\text{Hot } KMnO_4}$ **?**
Ans:
(i) $CH_3CH_2C{\equiv}CCH_3 \xrightarrow{\text{Cold dilute } KMnO_4} CH_3CH_2COCOCH_3$

(ii) $CH_3CH_2C{\equiv}CCH_3 \xrightarrow{\text{Hot } KMnO_4} CH_3CH_2COOH + CH_3COOH$

7. Give reason why chlorination of ethane to ethyl chloride is more practicable than chlorination of n-pentane to 1-chlorepentane.
Ans: Ethane forms only one monochloro derivative on chlorination while neopentane forms 3 isomers on monochlorination and the yield of 1-chloropentane is low as it is formed by replacement of primary hydrogen while other two isomers are formed by replacement of secondary hydrogen atoms respectively.

$CH_3CH_3 \xrightarrow{Cl_2} CH_3CH_2Cl$

$CH_3CH_2CH_2CH_2CH_3 \xrightarrow{Cl_2} CH_3(CH_2)_3CH_2Cl + CH_3CH(Cl)CH_2CH_2CH_3 + CH_3CH_2CH(Cl)CH_2CH_3$

8. How will you demonstrate that double bonds of benzene are somewhat different from that of olefins?
Ans: The double bonds of olefins decolourize bromine water and discharge the pink colour of Bayer's reagent while those of benzene not.

9. How will you separate propene from propyne?
Ans: By passing the mixture through ammonical silver nitrate solution when propyne reacts while propene passes over.

10. Write the structure of the alkene which on reductive ozonolysis gives butanone and ethanol.
Ans: $CH_3CH_2C(CH_3)=CHCH_3$

11. Explain the following name reactions with equations
(i) Wurtz reaction
(ii)Kolbe's electrolysis
(iii) Friedel- Crafts alkylation
(iv) Friedel-Crafts acylation
Ans:
(i) **Wurtz Reaction**

$$CH_3Br + 2Na + BrCH_3 \xrightarrow{\text{dry ether}} CH_3-CH_3 + 2NaBr$$

Bromomethane Ethane

(ii) **Kolbe's Electrolysis**

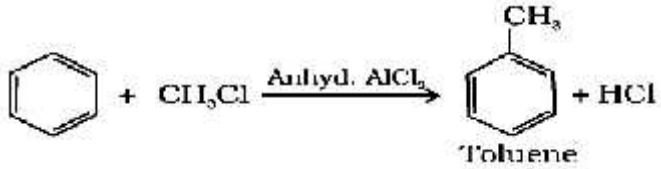
$$2CH_3COO^-Na^+ + 2H_2O$$

Sodium acetate

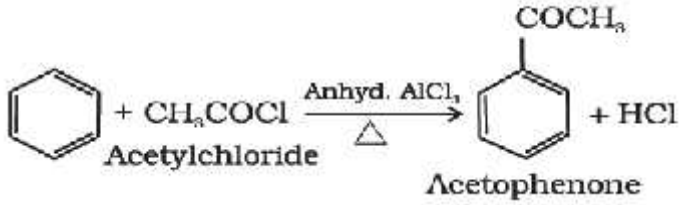
$$\downarrow \text{Electrolysis}$$

$$CH_3-CH_3 + 2CO_2 + H_2 + 2NaOH$$

(iii) Friedal Crafts Alkylation Reaction



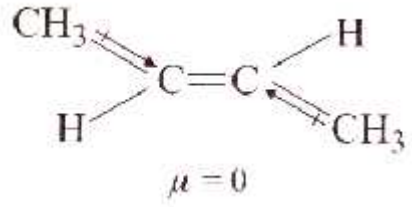
(iv) Friedal Crafts Acylation Reaction



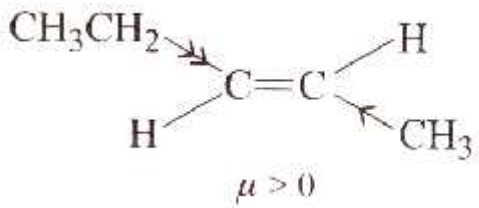
12. What are the main constituents of LPG?
Ans: The main constituents of LPG are butane and isobutane. But these isomers can be easily liquefied and hence can be conveniently transported in iron cylinders.

13. Is it possible to isolate pure staggered ethane or pure eclipsed ethane at room temperature? Explain.
Ans: The energy difference between the staggered and eclipsed form of ethane is just 12.55 $kJmol^{-1}$ which is easily met by collisions of the molecules at room temperature. Therefore, it is not possible to isolate either pure staggered or pure eclipsed ethane at room temperature.

14. *trans*-Pent-2-ene is polar while trans-but-2-ene is non polar. Explain.
Ans: In trans-but-2-ene, the dipole moments of the two C – CH_3 bonds are equal and hence they exactly cancel out each other. Thus trans-but-2-ene is non polar.



$\mu = 0$



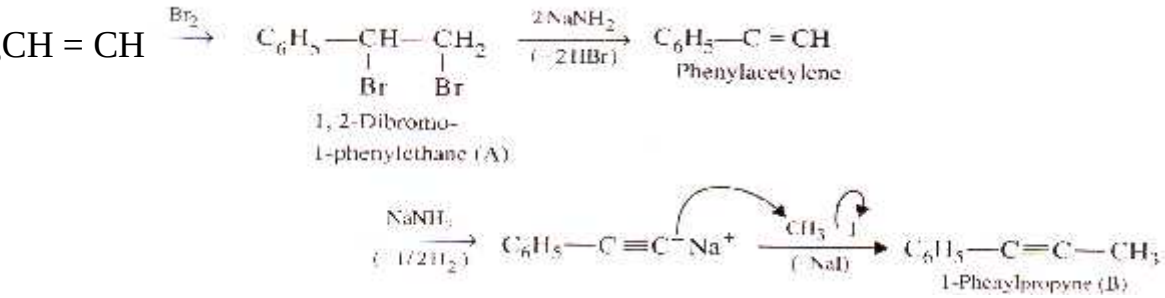
$\mu > 0$

However, in trans-Pent-2-ene, the +I effect of the CH_3CH_2 group is higher than that of CH_3 group, therefore the dipole moments of the C-CH_3 and C-CH_2CH_3 bonds are unequal. Although these dipoles, oppose each other, yet they do not exactly cancel out each other and hence, trans-pent-2-ene has a small but finite dipole moment and thus is polar.

15. Complete the following reactions with appropriate structure of products?

$$C_6H_5CH = CH \xrightarrow{Br_2} [A] \xrightarrow{(i)NaNH_2\ (3.0\ equiv)} B$$

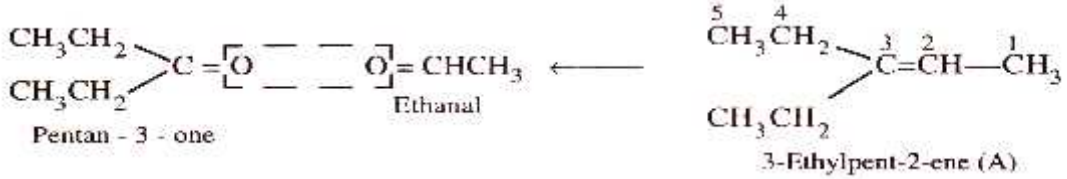
Ans:



Thus [A] is 1,2-dibromo-1-phenylethaneand [B] is 1-phenylpropyne.

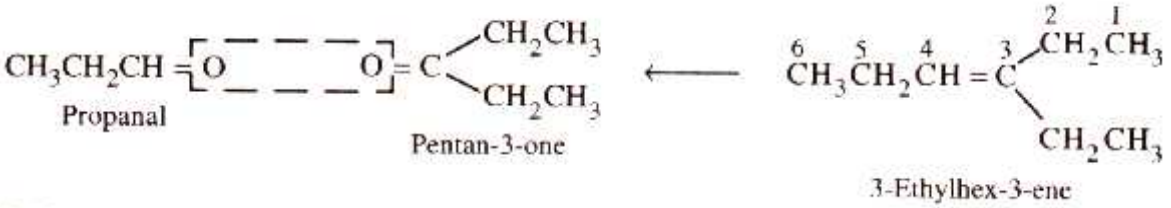
16. An alkene 'A' on ozonolysis gives mixture of ethano and pentan-3-one. Write the structure and IUPACX name of [A].

Ans: Write the wtructure o ft he products of the ozonolysis side by side with their oxygen atoms pointing towards each other. Remove the oxygen atoms and join the two ends by a double bond, the structure of the alkene [A] is



17. Propanol and pentanh-3-one are the ozonolysis products of an alkene. What is the structural formula of the Alkene?

Ans:



18. What are the favourable conditions for any system to be aromatic?

Ans: The necessary conditions for a moplecule to be aromatic are:

(v) It should have a single cyclic cloud of delocalized π- electrons above and below the plane of the molecule.

(vi) It should be planar. This is because the complete delocalization of the π electron is possible only if the ring is planar to allow cyclic overlap of p-orbitals.

(vii) It should contain Hückel number of electrons, i.e. (4n+2) π electrons where, n=0,1,2,3……. Etc.

19. What effect does branching of an alkene chain has on its boiling point?

Ans: As the branching increases, the surface are of an alkane approaches that of a sphere. Since, a sphere has minimum surface area, therefore, van der Waals forces of attraction are minimum and hence the boiling point of the alkane decreases with branching.

20. Why does benzene undergo electrophilic substitution reactions easily and neucleophilic substitution with difficulty?

Ans: Due to th presence of an electron cloud containing 6 π electro9ns above and below the plane of the ring, benzene is a rich source of electrons. Consequently, it attracts the electrophiles (electron deficient) reagenbts towards it and repels nuclephiles (electron rich) reagents. As a result benzene undergoes electrophilic substitution reactions easily and nucleophilic substitutions with difficulty.

21. Arrange the following set of compounds in order of their idecreasing relative reactivity withan electrophile E^+:

(a) Chlorobenzene, 2,4-dinitrochlorobenzene, p-nitrochlorobenzene.

(b) Toluene, p- $H_3C—C_6H_4—NO_2$, p-$O_2N—C_6H_4—NO_2$

Ans: The typical relations of benzene are electrophilic substitution reactions. Higher the electron density in the benzene ring, more reactive is the compound towards these reactions. Since, NO_2 is a more powerful electron – withdrawing group than Cl, more the number of two nitro groups, less reactive is the compound. Thus the overall reactivity decreases in the order:

Chlorobenzene > p-nitrochlorobenzene > 2,4-dinitrochlorobenzene

(b) Hence, CH3 group is electron donating but NO2 group is electron withdrawing. Therefore the maximum electron density will be in toluene, followed by p-nitrotoluene, followed by p-dinitrobenzene. Thus the overall reactivity decreases in the order:

Toluene > p- $H_3C—C_6H_4—NO_2$ > p-$O_2N—C_6H_4—NO_2$

22. an alkyl halide $C_5H_{11}Br$ (A) reacts with ethanolic KOH to give an alkene 'B', which reacts with Br_2 to give a compound C, which on dehydrobromination gives and alkyne 'D'. On treatment with Sodium Metal in liquid ammonia one mole of 'D' gives one mole of the sodium salt of 'D' and half a mole of hydrogen gas. Complete hydrogenation of 'D' yields a straight chain alkane. Identify A, B, C and D. Give the reactions involved.

Ans:

$$\underset{\text{Alkyl halide}}{\underset{(A)}{C_5H_{11}Br}} \xrightarrow[\Delta]{\text{alc. KOH}} \underset{\text{Alkene}}{\underset{(B)}{C_5H_{10}}} \xrightarrow[CS_2]{Br_2} \underset{\text{Dibromo alkane}}{\underset{(C)}{C_5H_{10}Br_2}}$$

$$\xrightarrow[\Delta]{\text{alc. KOH}} \underset{\text{Alkyne}}{\underset{(D)}{C_5H_8}} \xrightarrow{H_2} \underset{\text{(Straight chain alkane)}}{C_5H_{12}}$$

$$(D) \xrightarrow{\text{Na/liq. NH}_3} C_5H_7^-Na^+ + \frac{1}{2}H_2$$

Hydrogenation of alkyne (*D*) gives straight chain alkane hence all the compounds (*A*), (*B*), (*C*) and (*D*) must be straight chain compounds. Alkyne (*D*) forms sodium salt which proves that it is terminal alkyne. Involved reactions are given below :

$$\underset{\text{1-Bromopentane (A)}}{CH_3—CH_2CH_2CH_2—CH_2Br} \xrightarrow[-HBr]{\text{alc. KOH}}$$

$$\underset{\text{1-Pentene (B)}}{CH_3—CH_2CH_2CH=CH_2} \xrightarrow[CS_2]{Br_2}$$

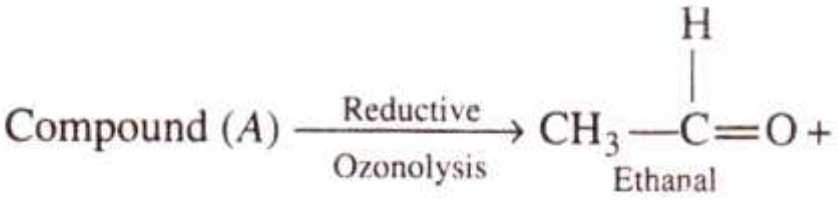
$$\underset{\text{1, 2-Dibromopentane (C)}}{CH_3—CH_2CH_2\overset{\overset{Br}{|}}{CH}—\overset{\overset{Br}{|}}{CH_2}} \xrightarrow[-2HBr]{\text{alc. KOH}}$$

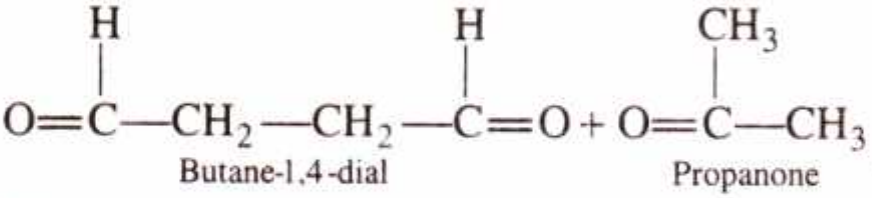
$$\underset{\text{1-Pentyne (D)}}{CH_3—CH_2CH_2C \equiv CH} \xrightarrow{H_2} CH_3CH_2CH_2CH_2CH_3$$

$$\xrightarrow{\text{Na/liq. NH}_3} CH_3CH_2CH_2C \equiv \overset{-}{C}\overset{+}{N}a + \frac{1}{2}H_2$$

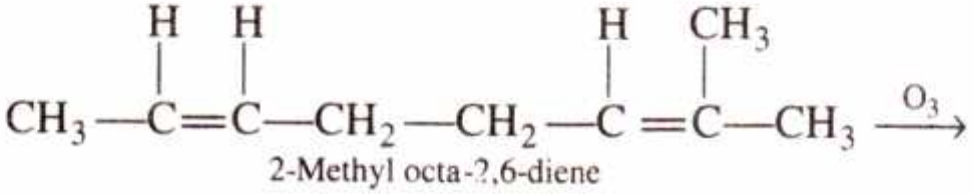
23. An unsaturated hydrocarbon 'A' ads two molecules of H_2 and on reductive ozonolysis gives butan-1,4-dial, ethanal and propanone. Give the structure of 'A', write its IUPAC name and explain the reactions involved.

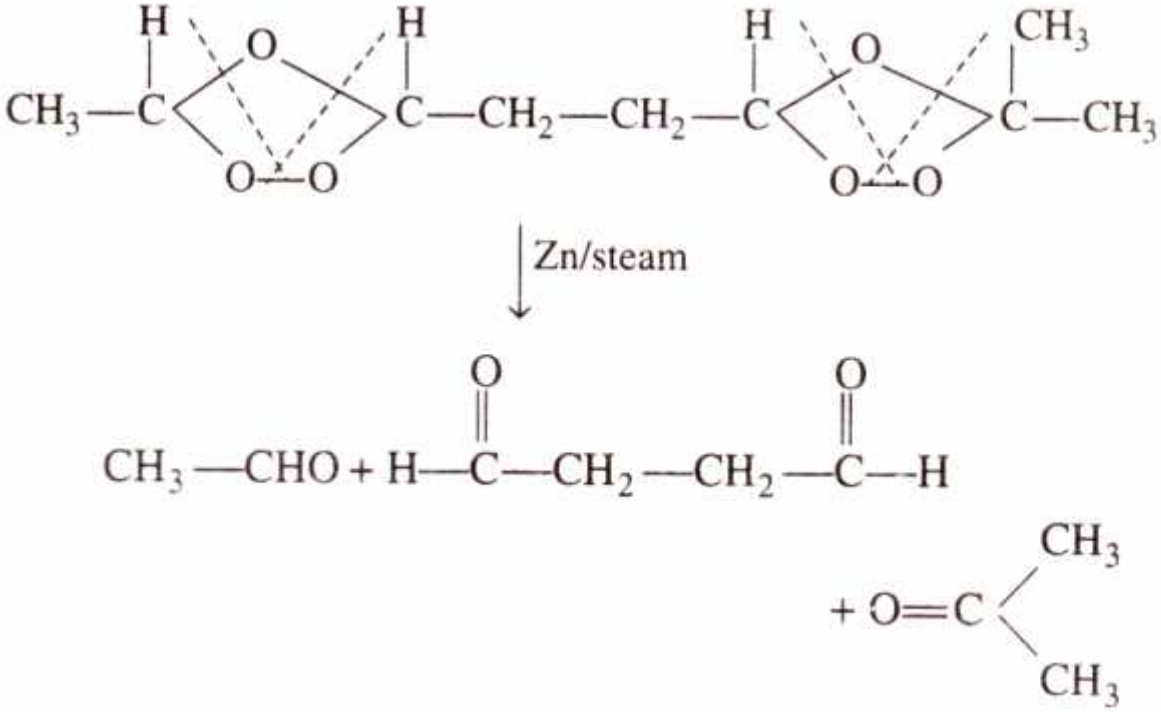
Ans:





Thus, structure of compound *A* may be given as :





24. In the presence of peroxide addition of HBr to propene takes place according top anti Markonikoff's rule but peroxide effect is not seen in the case of HCl and HI. Explain.

Ans:

$$H_3C—\underset{\text{Propene}}{CH=CH_2} + HBr \xrightarrow{\text{Peroxide}} H_3C—\underset{\text{n-Propyl bromide}}{CH_2CH_2Br}$$

Peroxide effect is effective only in the case of HBr and not seen in the case of HCl and HI. This is due to the following reasons :

(i) H—Cl bond (103 kcal/mol) is stronger than H—Br bond (87 kcal/mol).

H—Cl bond is not decomposed by the peroxide free radical, whereas the H—I bond is weaker (71 kcal/mol).

(ii) Iodine free radical ($I^\bullet$) formed as H—I bond is weaker but iodine free radicals readily combine with each other to form iodine molecules rather attacking the double bond.

VALUE BASED QUESTIONS

Please read the below narration and answer the questions that follow:

1. Kamala and Reema are best friends. On one evening, Kamala went to the house of Reema and found her in the kitchen. The gas burner was emitting yellow flame instead of blue flame. Kamala immediately asked Reema to put off the gas. She helped her in cleaning the fine holes of the gas burner with a needle. The process took about 15 minutes. The gas was ignited and there was a blue flame.

a) Why was the gas burner emitting yellow flame?

b) What was the purpose of cleaning the holes?

c) In what way Kamala helped Reema?

d) What lesson can we learn from this?

Answers

a) The holes of the burner were blocked due to soot or some oily material released during combustion. It was incomplete and the gas burnt with yellow flame.

b) By cleaning the holes, the deposits were removed and the combustion was now complete.

c) By cleaning the holes of the burner, there was a saving of fuel. This also helped in checking the pollution.

d) We should always keep gas burner and engines of scooters or cars clean so that the combustion is complete and there is no wastage of any fuel.

2. Teacher asked Hema to perform test for unsaturation in the laboratory for ethylene gas. She took some chlorine water in a tube and passed the vapours of the gas through it. Nothing happened. Teacher asked her to pass the vapours

of the gas into bromine water. The orange colour of bromine immediately discharged.

a) What was the mistake committed by Hema?

b) How did teacher help her?

c) Write chemical equation for the reaction.

Answers

a) Chlorine water has no colour. Therefore, on passing ethylene gas, the colour of chlorine water did not change.

b) Bromine water is orange in colour. When ethylene gas was passed through bromine water, its colour got discharged. This is the test for unsaturation.

c) $CH_2=CH_2 + Br_2 \rightarrow CH_2Br—CH_2Br$